Red, Falling, Nearest Star: Does Exposure to Science Fiction Literature Predict Knowledge and Processing of Genre-specific Collocations?

Daniel Jach

University of Jena

The 7th conference of the Scandinavian Association for Language and Cognition, Aarhus, Denmark
May 23, 2019

Introduction

There is good evidence from corpus-based psycholinguistic studies that language processing is sensitive to usage.

(Ellis, 2002; Gilquin & Gries, 2009)

Introduction

There is good evidence from corpus-based psycholinguistic studies that language processing is sensitive to usage.

(Ellis, 2002; Gilquin & Gries, 2009)

(1) sit in your chair *vs.* sit in your truck (Freq: high vs. low)

(Bannard & Matthews, 2008)

Introduction

There is good evidence from corpus-based psycholinguistic studies that language processing is sensitive to usage.

(Ellis, 2002; Gilquin & Gries, 2009)

(2) in one of the *vs.* in the context of the (Freq: high vs. low)

(3) a great deal of *vs.* is one of the (PMI: high vs. low)

(Ellis, Simpson-Vlach, & Maynard, 2008)

Introduction

There is good evidence from corpus-based psycholinguistic studies that language processing is sensitive to usage.

(Ellis, 2002; Gilquin & Gries, 2009)

(4) stinky baby *vs.* silly baby (Info(A): high vs. low)

(Bannard, Rosner, & Matthews, 2017)

Introduction

There is good evidence from corpus-based psycholinguistic studies that language processing is sensitive to usage.

(Ellis, 2002; Gilquin & Gries, 2009)

Language learning is a by-product of language use and therefore involves distributional knowledge. Recurrent strings are stored and processed as chunks.

Introduction

In a novel "words that go together" task, Dąbrowska (2014) found that the ability to recognize established collocations is independent of common corpus-based measures of association (e.g., mutual information, joint frequency).

(5)
delicate tea
feeble tea
frail tea
powerless tea
weak tea

(6)
deliver a speech
hold a speech
perform a speech
present a speech
utter a speech

Introduction

In a novel "words that go together" task, Dąbrowska (2014) found that the ability to recognize established collocations is independent of common corpus-based measures of association (e.g., mutual information, joint frequency).

(5)
- delicate tea
- feeble tea
- frail tea
- powerless tea
- X weak tea

(6)
- deliver a speech
- hold a speech
- perform a speech
- present a speech
- utter a speech

Introduction

In a novel "words that go together" task, Dąbrowska (2014) found that the ability to recognize established collocations is independent of common corpus-based measures of association (e.g., mutual information, joint frequency).

(5)
- delicate tea
- feeble tea
- frail tea
- powerless tea
- X weak tea

(6)
- X deliver a speech
- hold a speech
- perform a speech
- present a speech
- utter a speech

Introduction

In a novel "words that go together" task, Dąbrowska (2014) found that the ability to recognize established collocations is independent of common corpus-based measures of association (e.g., mutual information, joint frequency).

(5)
delicate tea
feeble tea
frail tea
powerless tea
X weak tea

(6)
X deliver a speech
hold a speech
perform a speech
present a speech
utter a speech

This is odd! If learning is usage-based and corpora represent usage, then corpus-based measures of association should be indices and drivers of emergent linguistic chunks.

Introduction

In a novel "words that go together" task, Dąbrowska (2014) found that the ability to recognize established collocations is independent of common corpus-based measures of association (e.g., mutual information, joint frequency).

(5)
- delicate tea
- feeble tea
- frail tea
- powerless tea
- X weak tea

(6)
- X deliver a speech
- hold a speech
- perform a speech
- present a speech
- utter a speech

This is odd! If learning is usage-based and corpora represent usage, then corpus-based measures of association should be indices and drivers of emergent linguistic chunks.

Instead, task performance was significantly correlated with print exposure (Author Recognition Test (ART), Acheson, Wells, & MacDonald, 2008).

The Current Study

Question What is the relationship between speakers' print exposure, corpus-based measures of association, and speakers' use and processing of collocations?

Method A bigram completion task was conducted to determine the effect of print exposure to English science fiction (SF) literature and corpus-based measures of association on readers' responses.

Results Responses at higher levels of print exposure reflected the usage distribution of bigrams in SF literature more closely.

Overview

1. Introduction
2. Method
3. Results
4. Discussion

Section 2

Method

The Science Fiction Corpus (SFC)

Corpus

- 1,548 English science fiction texts from Gutenberg.org
- 32,402,850 words, 0.01 type/token ratio, 16.49 token/sentence ratio
- POS tagging with TreeTagger with BNC tagset for Linux

Rank	N	A	RS authors	RS titles
1	man	other	H.G. Wells	The Case of Summerfield
2	time	great	Zona Gale	The Infra-Medians
3	way	little	Murray Leinster	Postmark Ganymede
4	men	good	H. Beam Piper	The Calm Man
5	eyes	old	John Joseph McGuire	Perfect Control
6	hand	long	Ron Cocking	The Buttoned Sky
7	face	new	Fritz Leiber	Venus Boy
8	world	small	Henry Slesar	Syndrome Johnny
9	people	sure	C.L. Moore	Fifteen Hundred Miles An Hour
10	earth	right	Mike Lewis	Master of Life and Death

Corpus-based Measures

- 593,873 unique adjective-noun bigrams
- Joint frequency, mutual information, informativity of adjective

AN	Rank	BNC	PMI(AN)	BNC	Info(A)	BNC
old man	1	21	4.40	4.34	4.09	4.61
young man	2	18	4.98	5.01	4.23	4.43
long time	3	4	4.15	4.43	4.53	5.16
other side	4	12	4.57	4.29	3.32	3.38
extensive research	5	3890	9.23	4.33	1.01	8.31
other hand	6	2	3.77	4.87	4.47	2.53
solar system	7	1573	7.64	5.58	2.53	8.41
great deal	8	6	6.06	6.89	1.26	1.19
human beings	9	49	7.16	8.74	1.16	0.41
whole thing	10	74	4.75	5.06	4.71	4.96

Author Recognition Test (Print Exposure)

- Author recognition test to assess participants' SF print exposure (Acheson et al., 2008; Stanovich & West, 1989)
- 16 SFC author names, 16 fake names (Martin-Chang & Gould, 2008)
- Participants indicate authors they know, penalty for fakes

Author Recognition Test (Print Exposure)

- Author recognition test to assess participants' SF print exposure (Acheson et al., 2008; Stanovich & West, 1989)
- 16 SFC author names, 16 fake names (Martin-Chang & Gould, 2008)
- Participants indicate authors they know, penalty for fakes

Please click on the authors you recognize. Careful! I have added some fake names to make sure you are not just guessing.

Alex James	☐	...	☐
Andrew Lang	☐	Richard Wilson	☐
Arthur Conan Doyle	☐	Robert Louis Stevenson	☐
Ayn Rand	☐	Rudyard Kipling	☐
...	☐	Thomas Bever	☐

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ star

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ star

red

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ star

falling

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ star

nearest

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ sun

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ sun

flaring

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ sun

blazing

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ sun

collapsing

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ animal

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ animal

savage

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ animal

friendly

Bigram Completion Task (BCT)

- 3,398 genre-specific bigrams (collocations in SFC but not in BNC)
  - 100 noun types with high standard deviation in both joint frequency and mutual information

_____ animal

dangerous

Procedure and Predictions

Procedure

- Ibex farm (Drummond, 2013)
- Participant recruitment online (callforparticipants.com, PRO, SFWA, r/SampleSize)
- Instruction $\rightarrow$ ART $\rightarrow$ 3 practice trials $\rightarrow$ BCT $\rightarrow$ Lottery
- Each participant completed random sample of 50 nouns
- Response (adjective) and response time (RT) were recorded

Procedure and Predictions

Procedure

- Ibex farm (Drummond, 2013)
- Participant recruitment online (callforparticipants.com, PRO, SFWA, r/SampleSize)
- Instruction → ART → 3 practice trials → BCT → Lottery
- Each participant completed random sample of 50 nouns
- Response (adjective) and response time (RT) were recorded

Predictions

1. Participants are sensitive to usage distributions.
2. More experienced SF readers produce more strongly associated and more informative bigrams.
3. More experienced SF readers produce more genre-specific bigrams (collocations).
4. RT decreases with increasing association strength.

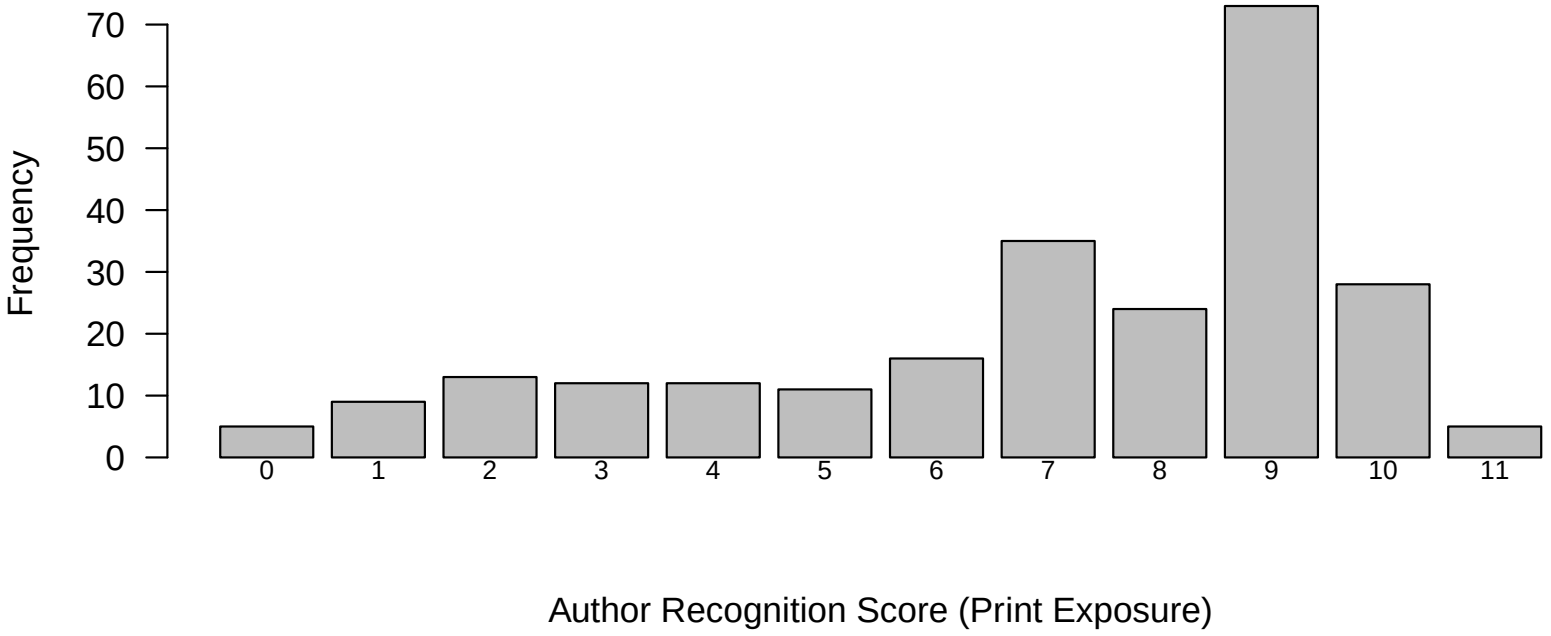
Section 3

Results

Results

Table: Summary of participants' background information

Variable	
N	243
Sex (female): %	59.26
Age in years: *M* (*SD*)	38.33 (15.35)
L1	English
ART score (-16 – 16): *M* (*SD*)	6.99 (2.8)



Joint Frequency Distribution

Prediction

1. Participants are sensitive to usage distributions.

Joint Frequency Distribution

Prediction

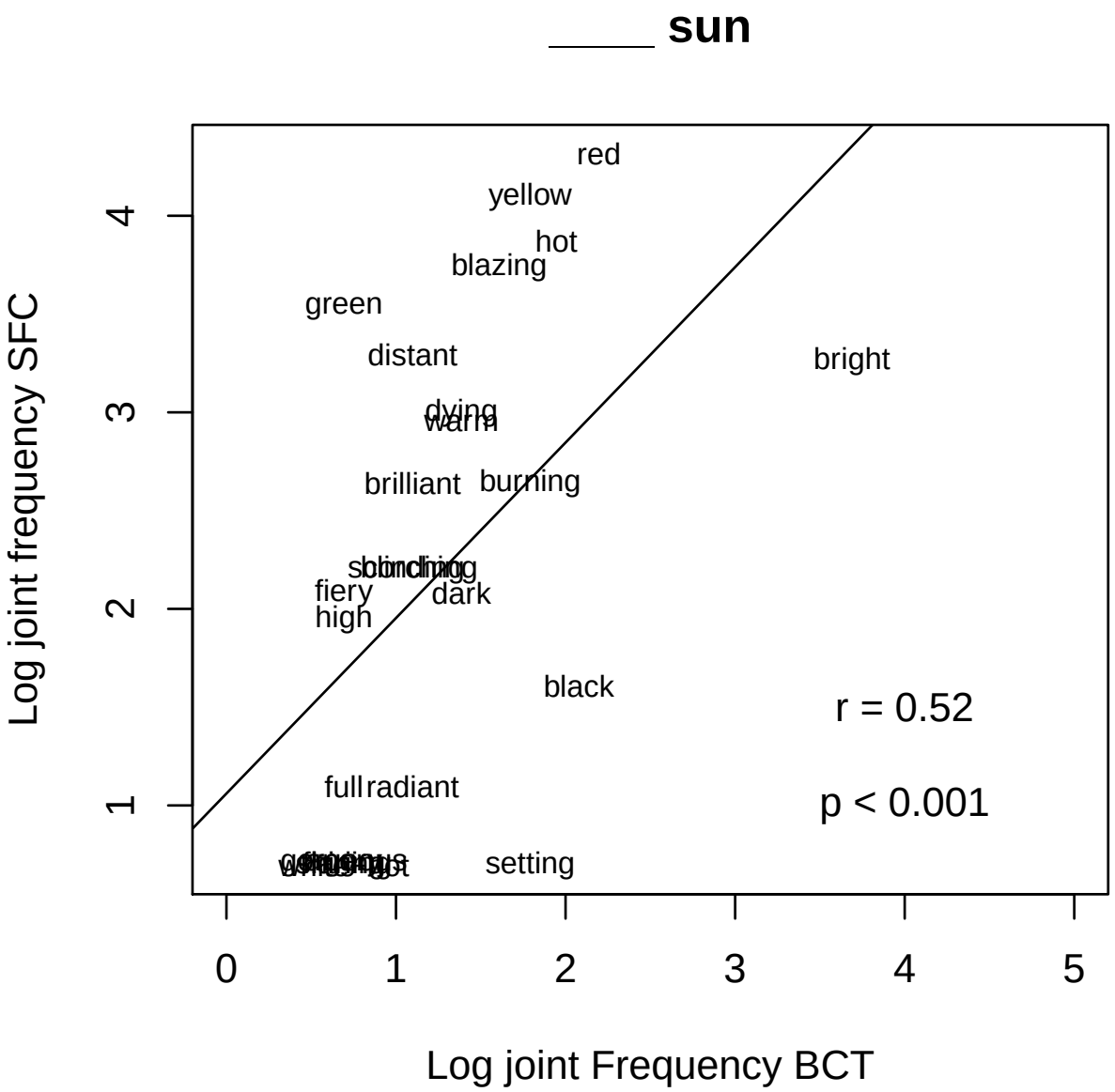
1. Participants are sensitive to usage distributions.

For each noun:

Frequency of bigram elicitation $\sim$ Frequency of bigram use in SFC/BNC

(Ellis, Römer, & O'Donnell, 2016, pp. 103–107)

Joint Frequency Distribution



Joint Frequency Distribution

a Correlations BCT $\sim$ SFC

N	n(A)	r SFC	p
arm	18	0.82	***
safety	13	0.69	**
direction	22	0.60	**
war	29	0.51	**
music	19	0.42	.
voice	42	0.37	*
circumstances	30	0.33	.
condition	28	0.25	ns
building	27	0.16	ns
star	21	0.03	ns

M of r (n of $p \leq .05$): 0.37 (40)

Joint Frequency Distribution

a Correlations BCT $\sim$ SFC

N	n(A)	r SFC	p
arm	18	0.82	***
safety	13	0.69	**
direction	22	0.60	**
war	29	0.51	**
music	19	0.42	.
voice	42	0.37	*
circumstances	30	0.33	.
condition	28	0.25	ns
building	27	0.16	ns
star	21	0.03	ns

M of r (n of $p \leq .05$): 0.37 (40)

b Correlations BCT $\sim$ BNC

N	n(A)	r BNC	p
term	8	0.91	**
event	22	0.67	***
clothes	32	0.56	***
discovery	28	0.51	**
spirit	23	0.46	*
device	23	0.40	.
office	28	0.35	.
system	21	0.29	ns
form	26	0.23	ns
path	33	0.13	ns

M of r (n of $p \leq .05$): 0.39 (44)

PMI(AN) and Info(A) Distribution

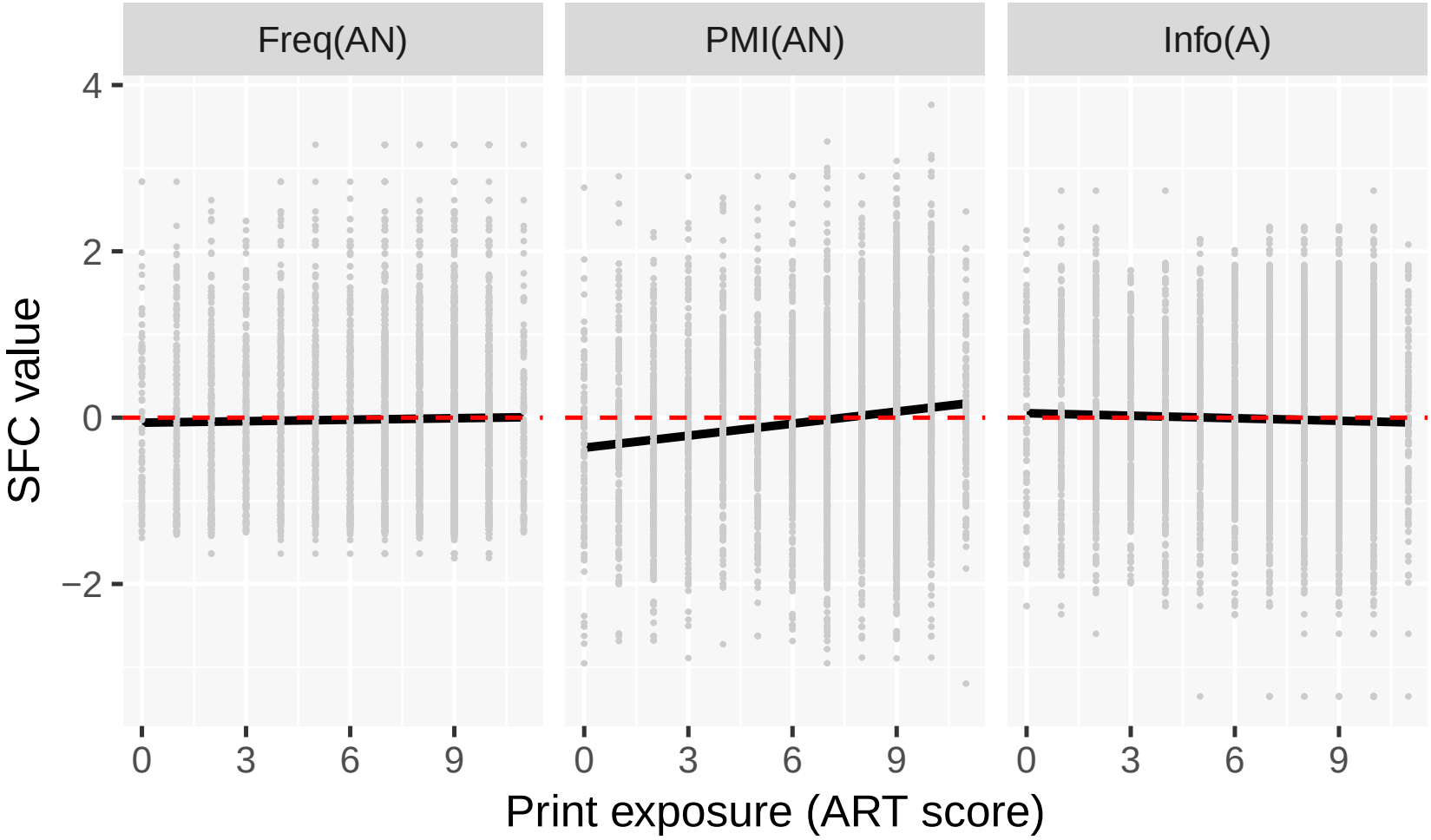
Prediction

2. More experienced SF readers produce more strongly associated and more informative bigrams.

PMI(AN) and Info(A) Distribution

Prediction

2. PMI(AN), Info(A) $\sim$ Print exposure (ART score)



	b	*SE*	*t*	*p*
Intercept	-0.06	0.04	-1.59	ns
Print exposure (ART score)	0.01	0	1.29	ns
PMI(AN)	-0.3	0.05	-6.14	***
Info(A)	0.11	0.05	2.31	*
Print exposure (ART score) x PMI(AN)	0.04	0.01	6.57	***
Print exposure (ART score) x Info(A)	-0.02	0.01	-2.48	*
Number of observations	6269			

Genre-specific Bigrams (Collocations)

Prediction

3. More experienced SF readers produce more genre-specific bigrams.

Genre-specific Bigrams (Collocations)

Prediction

3. SFColl(yes/no) $\sim$ Print exposure (ART score)

Genre-specific Bigrams (Collocations)

Prediction

3. SFColl(yes/no) $\sim$ Print exposure (ART score)

	b	*SE*	*z*	*p*
Intercept	-2.62	0.12	-21.21	***
Print exposure (ART score)	0.03	0.01	2.85	**
Number of observations	12150			

Response Time (RT)

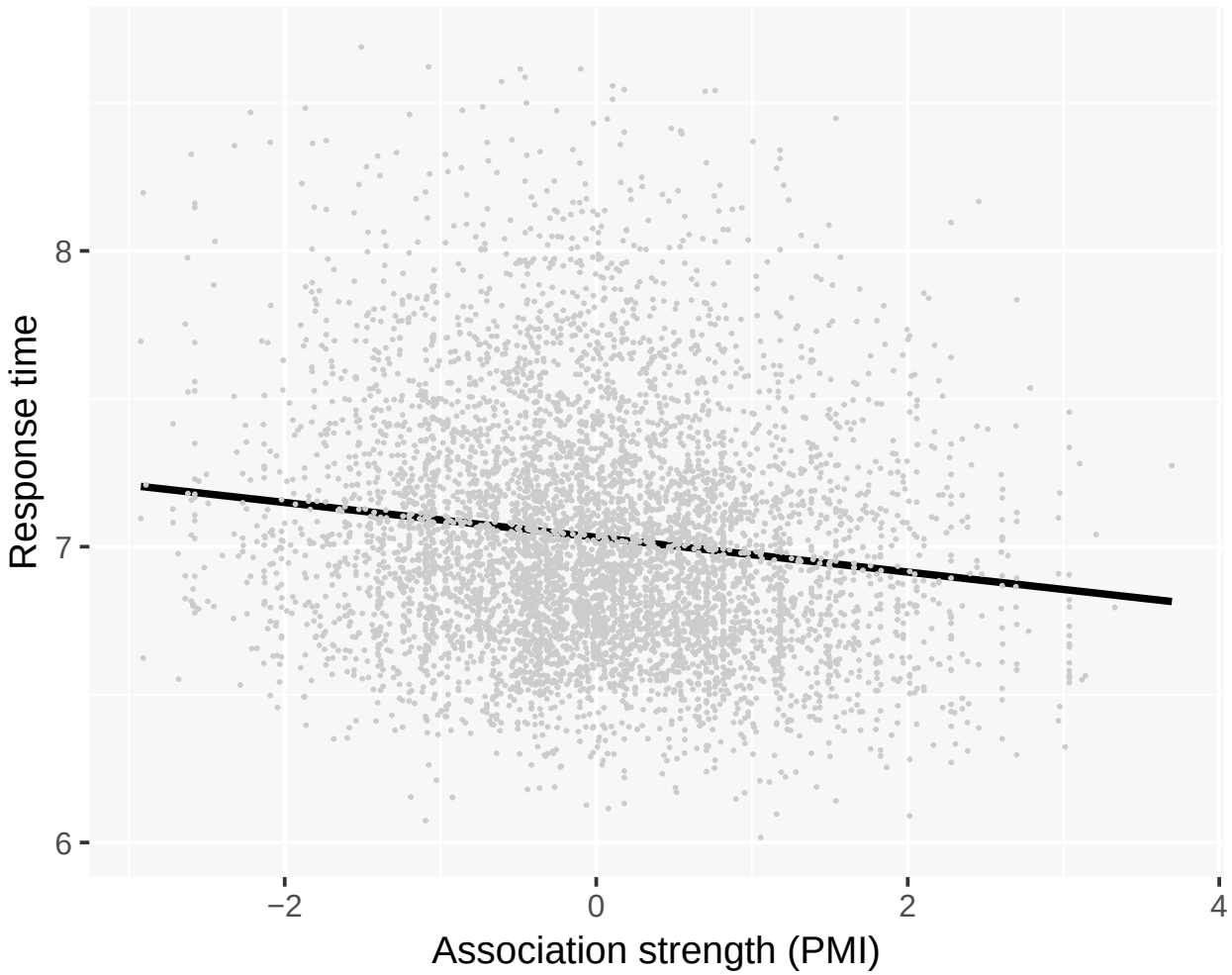
Prediction

4. RT decreases with increasing association strength.

Response Time (RT)

Prediction

4. $\text{RT} \sim \text{Freq(AN)} + \text{PMI(AN)}$



	b	*SE*	*t*	*p*
Intercept	7.008	0.049	143.885	***
Freq	0	0.01	0.02	ns
PMI	-0.059	0.01	-5.832	***
Print exposure (ART score)	0.01	0.006	1.599	ns
Trial	-0.002	0	-4.431	***
Number of observations	5982			

Section 4

Discussion

Discussion

Results

- The participants were sensitive to the usage distribution of bigrams.
- With increasing print exposure, SF readers produced more strongly associated and more genre-specific bigrams.
- Respone time decreased with increasing association strength.

Discussion

Results

- The participants were sensitive to the usage distribution of bigrams.
- With increasing print exposure, SF readers produced more strongly associated and more genre-specific bigrams.
- Respone time decreased with increasing association strength.

Language learning is a by-product of language use and therefore involves distributional knowledge. Recurrent strings are stored and processed as chunks.

(Diessel, 2016; Ellis et al., 2016; Schmid, 2018; Sinclair, 1991)

Discussion

Results

- The participants were sensitive to the usage distribution of bigrams.
- With increasing print exposure, SF readers produced more strongly associated and more genre-specific bigrams.
- Respone time decreased with increasing association strength.

With accumulating experience, the effect of high joint frequency on learning levels out because once a routine has been established additional practice brings little improvement; instead, language users fine-tune their constructions to more specific contexts of use.

(Durrant & Schmitt, 2009; Ellis et al., 2008; Kaan, 2014; Jach, 2018; Schmitt, 2012)

Limitations and Outlook

Limitations

- Unexpected, weak effect of informativity on responses
- No interaction Print exposure $\times$ Mutual information $\times$ RT

Outlook

- Mismatch between BCT and SFC bigrams
- Bi-, tri-, quadro-, n-grams

Thank you for your attention.

References I

Acheson, D. J., Wells, J. B., & MacDonald, M. C. (2008). New and updated tests of print exposure and reading abilities in college students. *Behavior Research Methods*, *40*(1), 278–289. doi:10.3758/BRM.40.1.278

Bannard, C. & Matthews, D. (2008). Stored word sequences in language learning: The effect of familiarity on children's repetition of four-word combinations. *Psychological Science*, *19*(3), 241–248. doi:10.1111/j.1467-9280.2008.02075.x. eprint: https://doi.org/10.1111/j.1467-9280.2008.02075.x

Bannard, C., Rosner, M., & Matthews, D. (2017). What's worth talking about? information theory reveals how children balance informativeness and ease of production. *Psychological Science*, *28*(7), 954–966. PMID: 28598257. doi:10.1177/0956797617699848. eprint: https://doi.org/10.1177/0956797617699848

Dąbrowska, E. (2014). Words that go together: Measuring individual differences in native speakers' knowledge of collocations. *The Mental Lexicon*, *9*(3), 401–418. doi:10.1075/ml.9.3.02dab

References II

Diessel, H. (2016). Frequency and lexical specificity in grammar: A critical review. In H. Behrens & S. Pfänder (Eds.), *Experience counts: Frequency effects in language* (pp. 209–238). Berlin, Germany: De Gruyter.

Drummond, A. (2013). Ibex farm. Retrieved February 12, 2017, from http://spellout.net/ibexfarm.

Durrant, P. & Schmitt, N. (2009). To what extent do native and non-native writers make use of collocations? *International Review of Applied Linguistics in Language Teaching*, *47*(2), 157–177. doi:10.1515/iral.2009.007

Ellis, N. C. (2002). Frequency effects in language processing: A review with implications for theories of implicit and explicit language acquisition. *Studies in Second Language Acquisition*, *24*(2), 143–188. doi:10.1017/s0272263102002024

Ellis, N. C., Römer, U., & O'Donnell, M. B. (2016). *Usage-based approaches to language acquisition and processing: Cognitive and corpus investigations of construction grammar*. Malden, MA: Wiley-Blackwell.

References III

- Ellis, N. C., Simpson-Vlach, R., & Maynard, C. (2008). Formulaic language in native and second language speakers: Psycholinguistic, corpus linguistics, and TESOL. *TESOL Quarterly*, *42*(3), 375–396. doi:10.1002/j.1545-7249.2008.tb00137.x
- Gilquin, G. & Gries, S. T. (2009). Corpora and experimental methods: A state-of-the-art review. *Corpus Linguistics and Linguistic Theory*, *5*(1), 1–26. doi:10.1515/cllt.2009.001
- Jach, D. (2018). A usage-based approach to preposition placement in English as a second language. *Language Learning*, *68*(1), 271–304. doi:10.1111/lang.12277
- Kaan, E. (2014). Predictive sentence processing in L2 and L1: What is different? *Linguistic Approaches to Bilingualism*, *4*(2), 257–282. doi:10.1075/lab.4.2.05kaa

References IV

Martin-Chang, S. L. & Gould, O. N. (2008). Revisiting print exposure: Exploring differential links to vocabulary, comprehension and reading rate. *Journal of Research in Reading*, *31*(3), 273–284. doi:10.1111/j.1467-9817.2008.00371.x. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-9817.2008.00371.x

Schmid, H.-J. (2018). Unifying entrenched tokens and schematized types as routinized commonalities of linguistic experience. In *Yearbook of the german cognitive linguistics association* (Vol. 6, *1*, pp. 167–182). doi:10.1515/gcla-2018-0008

Schmitt, N. (2012). Formulaic language and collocation. In C. A. Chapelle (Ed.), *The encyclopedia of applied linguistics* (pp. 1–10). Malden, MA: Wiley-Blackwell. doi:10.1002/9781405198431.wbeal0433

Sinclair, J. (1991). *Corpus, concordance, collocation*. Oxford, England: Oxford University Press.

References V

Stanovich, K. E. & West, R. F. (1989). Exposure to print and orthographic processing. *Reading Research Quarterly*, *24*(4), 402–433. Retrieved from http://www.jstor.org/stable/747605